

Part G: Scan Report

Scan Date

28 June 2026

Target System

Localhost (127.0.0.1) – Kali Linux Virtual Machine

Commands Used

nmap localhost

nmap -sV localhost

sudo nmap -O localhost

Open Ports

Port	Status
None	No open ports found

Running Services

Port	Service	Version
None	None	None

Operating System

Detection	Result
Operating System	Not detected

Reason:

Nmap could not identify the operating system because all scanned ports were closed, resulting in insufficient information for accurate OS fingerprinting.

Observations

- The localhost system was reachable.

- No open TCP ports were found.
- No network services were running.
- Service version detection returned no results because there were no open ports.
- Operating system detection was inconclusive due to the lack of open ports.
- The system presents a small network attack surface since no services are exposed.

Recommendations

- Keep only required services running.
- Regularly scan the system to detect unexpected open ports.
- Keep the operating system and applications updated.
- Enable a firewall to restrict unauthorized network access.
- Use strong passwords and secure configurations for any services that may be enabled in the future.

Conclusion

The scanning phase demonstrated how Nmap can be used to assess the security posture of a system by identifying open ports, running services, and operating system information. During this exercise, the localhost system had no open TCP ports, so no active services or service versions were detected. Operating system detection was also unsuccessful because Nmap requires responses from open or filtered ports to accurately identify the OS. This exercise showed that a system with no unnecessary network services has a smaller attack surface and is generally more secure against network-based attacks. I also learned how to perform basic port scanning, service version detection, and OS detection using Nmap. Understanding these scanning techniques is important because they help security professionals identify exposed services, verify system configurations, and detect potential security risks before attackers can exploit them. Overall, the activity provided practical experience with network reconnaissance while emphasizing the importance of maintaining a secure and well-configured system.